

2. Nuclides containing even numbers of both protons and neutrons are most stable means less radioactive.
3. Nuclides contain odd numbers of protons and even numbers of neutrons are less stable than nuclides containing even numbers of protons and odd numbers of neutrons.

In general, nuclear stability is greater for nuclides containing even numbers of protons and neutrons or both.

Rates of Radioactive Decay

Radioactive Half-Life:

The radioactive half-life for a given radioisotope is a measure of the tendency of the nucleus to "decay" or "disintegrate" and as such is based purely upon that probability. The tiny nuclear size compared to the atom and the enormity of the forces which act within it make it almost totally impervious to the outside world. The half-life is independent of the physical state (solid, liquid, gas), temperature, pressure, the chemical compound in which the nucleus finds itself, and essentially any other outside influence. It is independent of the chemistry of the atomic surface, and independent of the ordinary physical factors of the outside world. The only thing which can alter the half-life is direct nuclear interaction with a particle from outside, e.g., a high energy collision in an accelerator.

The predictions of decay can be stated in terms of the half-life, the decay constant, or the average lifetime. The relationship between these quantities is as follows.

$$T_{1/2} = \frac{.693}{\lambda} = .693 \tau$$

λ ← Radioactive decay constant.

Nuclear Decay Probability:

Radioactive decay is a statistical process which depends upon the instability of the particular radioisotope, but which for any given nucleus in a sample is completely unpredictable. The decay process and the observed half-life dependence of radioactivity can be predicted by assuming that individual nuclear decays are purely random events. If there are N radioactive nuclei at some time t , then the number ΔN which would decay in any given time interval Δt would be proportional to N :

The radioactive half life for a given radioisotope is the time for half the radioactive nuclei in any sample to undergo radioactive decay. After two half-lives, there will be one fourth the original sample, after three half-lives one eighth the original sample and so forth.